



Repsol Eco-Fuels Demand Forecasting Project

Macro Demand Forecast for Diesel Nexa / Biodiesel in Spain

Forecast horizon: January 2026 to December 2027

Geographic scope: Spain, Madrid, Catalonia, Andalusia, and Valencia

Final report draft | 29 June 2026

Ethics statement: This report uses public or project-provided datasets, avoids personal data, documents uncertainty transparently, and presents forecasts as decision-support estimates rather than deterministic claims.

AI disclosure: AI tools supported drafting, structuring, formatting, and quality review. Analytical outputs and interpretations were checked against repository artifacts and validation scripts.



Executive Summary

This project develops a monthly demand forecasting framework for Diesel Nexa / biodiesel demand in the Spanish market. The aim is to estimate demand over a 24-month horizon, from January 2026 to December 2027, at both national and regional levels.

The project combines several data sources into a single analytical pipeline: biodiesel consumption, macroeconomic indicators, Brent oil prices, fuel retail prices, and CNMC petroleum context variables. These sources are consolidated into a monthly master dataset using `Fecha` and `CCAA` as the primary keys.

The final modeling framework evaluates statistical time-series models, growth-curve models, and machine-learning models. The selected model is chosen separately for each target series because biodiesel adoption does not evolve uniformly across Spain.

The final selected models are:

Target	Selected Model	MAE	RMSE	MAPE	R2
Nacional	Logistic	6,228.7	6,999.4	36.7%	-1.041
Madrid	Logistic	1,924.0	2,057.2	73.6%	-8.273
Cataluña	SARIMA	1,557.8	1,794.1	50.1%	-7.182
Andalucía	SARIMA	898.8	1,077.9	52.6%	-1.929
Valencia	Gompertz	360.3	446.6	34.2%	-1.246

The forecasts suggest continued national demand growth through 2026 and 2027. However, the results must be interpreted with caution because the available time series is short and the biodiesel market appears to be in an adoption-stage regime rather than a mature stable-demand regime.

1. Objective

1.1 Problem Definition

Repsol operates in an energy market shaped by decarbonization, regulatory pressure, fuel substitution, and changing customer expectations. Eco-fuels such as Diesel Nexa are strategically important because they may support lower-carbon mobility while using existing liquid-fuel infrastructure.

The business problem is that Diesel Nexa / biodiesel demand is difficult to forecast with traditional methods. The available data shows strong growth from a limited historical base, and the market does not yet behave like a mature fuel category. This creates uncertainty for commercial planning, supply preparation, and regional prioritization.

The core forecasting challenge is therefore:

How can monthly Diesel Nexa / biodiesel demand be forecasted for Spain and selected regions over the next 24 months using the available historical, macroeconomic, energy-price, and petroleum-context data?

1.2 Aim of the Work

The aim of the project is to build a reproducible forecasting pipeline that:

- Consolidates all relevant data sources into a consistent monthly analytical dataset.
- Produces model-ready features for national and regional demand forecasting.
- Evaluates multiple forecasting approaches rather than relying on a single algorithm.
- Selects the most appropriate model for each target market.
- Generates 24-month forecasts for 2026 and 2027.
- Communicates results in a way that is useful for both technical and business audiences.

1.3 Forecasting Scope

The forecasting scope includes five target series:

- Spain as a national market.
- Madrid.
- Catalonia.
- Andalusia.
- Valencia.

The final forecast horizon is monthly from 2026-01 to 2027-12.

1.4 Target Variable

The dependent variable is `Consumo_Tm`, measured in metric tons. In this project, it represents distinct biodiesel consumption reported as its own product line. It should not be interpreted as the total biodiesel fraction blended into all diesel products.

This definition matters because a blended biodiesel-content metric would describe renewable content across conventional diesel. The project target is closer to a product adoption and commercialization problem.

1.5 Expected Use of the Forecast

The forecast is intended to support:

- Medium-term commercial planning.
- Regional demand monitoring.
- Strategic supply and capacity discussions.
- Scenario interpretation for energy and fuel-price context.
- Identification of future data needs.

The forecast is not intended to be used alone for:

- Daily operational scheduling.
- Contract-level volume commitments.
- Causal policy evaluation.
- Pricing optimization without additional modeling.

2. Data Sources

2.1 Overview

The project uses multiple data sources to represent demand, macroeconomic context, energy-market context, and petroleum-market context. The sources have different original granularities, so preparation and alignment are required before modeling.

Data Source	Main File / Output	Role in the Project
CORES / biodiesel consumption	data/inputs/consumo_biodiesel_ccaa.csv	Core monthly demand series by autonomous community.
Target demand series	data/inputs/consumo_biodiesel_targets.csv	Modeling targets for Spain, Madrid, Catalonia, Andalusia, and Valencia.
INE macroeconomic indicators	data/inputs/macro_indicadores_ine.csv	Industrial production, inflation, and unemployment context.
Brent oil prices	data/inputs/brent_oil_price_monthly_2023_onwards.csv	International crude oil price context.
Fuel retail prices	data/inputs/precios_combustibles_2023.csv, 2024.csv, 2025.csv	Regional gasoline and diesel price signals.
CNMC petroleum context	Integrated in data/inputs/master_dataset.csv	Biodiesel reconciliation and diesel-pool context.

2.2 Consumption Data

The consumption data is the core of the forecasting problem. It provides monthly biodiesel demand by autonomous community. This dataset is used to create the national and regional target series.

Main columns:

- Fecha
- CCAA
- Consumo_Tm

The monthly structure allows the project to model trend, seasonality, lags, and rolling demand behavior.

2.3 Macroeconomic Data

The macroeconomic data comes from INE indicators and includes variables such as:

- IPI_original
- IPI_ajustado
- IPC_var_anual
- Tasa_paro

These variables provide economic context. Industrial production may reflect economic activity, inflation may affect consumption conditions, and unemployment may capture broader economic pressure.

2.4 Energy and Fuel Price Data

The project includes Brent oil prices and fuel retail prices. Fuel prices are important because biodiesel adoption may be influenced by relative fuel economics, customer substitution behavior, and broader energy-market conditions.

The fuel-price files are originally more granular than the modeling table. They are processed and aggregated into monthly variables by autonomous community.

Examples of final price-related columns include:

- Precio_Brent_USD
- PVP_Gasoleo_A

- PVP_Gasoleo_Premium
- PVP_Gasolina95
- PVP_Gasolina98
- PAI_Gasoleo_A
- PAI_Gasolina95

2.5 CNMC Petroleum Context

CNMC petroleum data provides additional context around biodiesel and diesel-pool behavior. This is useful because Diesel Nexa / biodiesel demand should be interpreted in relation to the broader diesel market.

Examples of integrated variables include:

- CNMC_Biodiesel_Tm
- GasoleoA_Tm
- DieselPool_Tm
- Ratio variables connecting biodiesel to the broader diesel pool.

2.6 Master Dataset Construction

The final master dataset is saved as:

`data/inputs/master_dataset.csv`

Current dimensions:

- 720 rows.
- 22 columns.
- 20 autonomous communities / national entities.
- Monthly coverage from 2023-01 to 2025-12.

The master dataset uses two primary keys:

- Fecha
- CCAA

Each row represents one month and one autonomous community. This design allows the same dataset to support national analysis, regional analysis, feature engineering, and model training.

2.7 Data Preparation and Processing

The main preparation steps are:

1. Clean and standardize biodiesel consumption data.
2. Normalize date columns into a monthly YYYY-MM format.
3. Standardize region names in CCAA.
4. Aggregate fuel prices from granular raw files into monthly regional values.
5. Align macroeconomic indicators by month.
6. Merge all compatible sources using Fecha and CCAA.
7. Preserve missing values where a data source cannot fully cover all regions or months.

The decision to preserve missing values is important. Dropping rows would reduce the already short time series and could bias regional coverage.

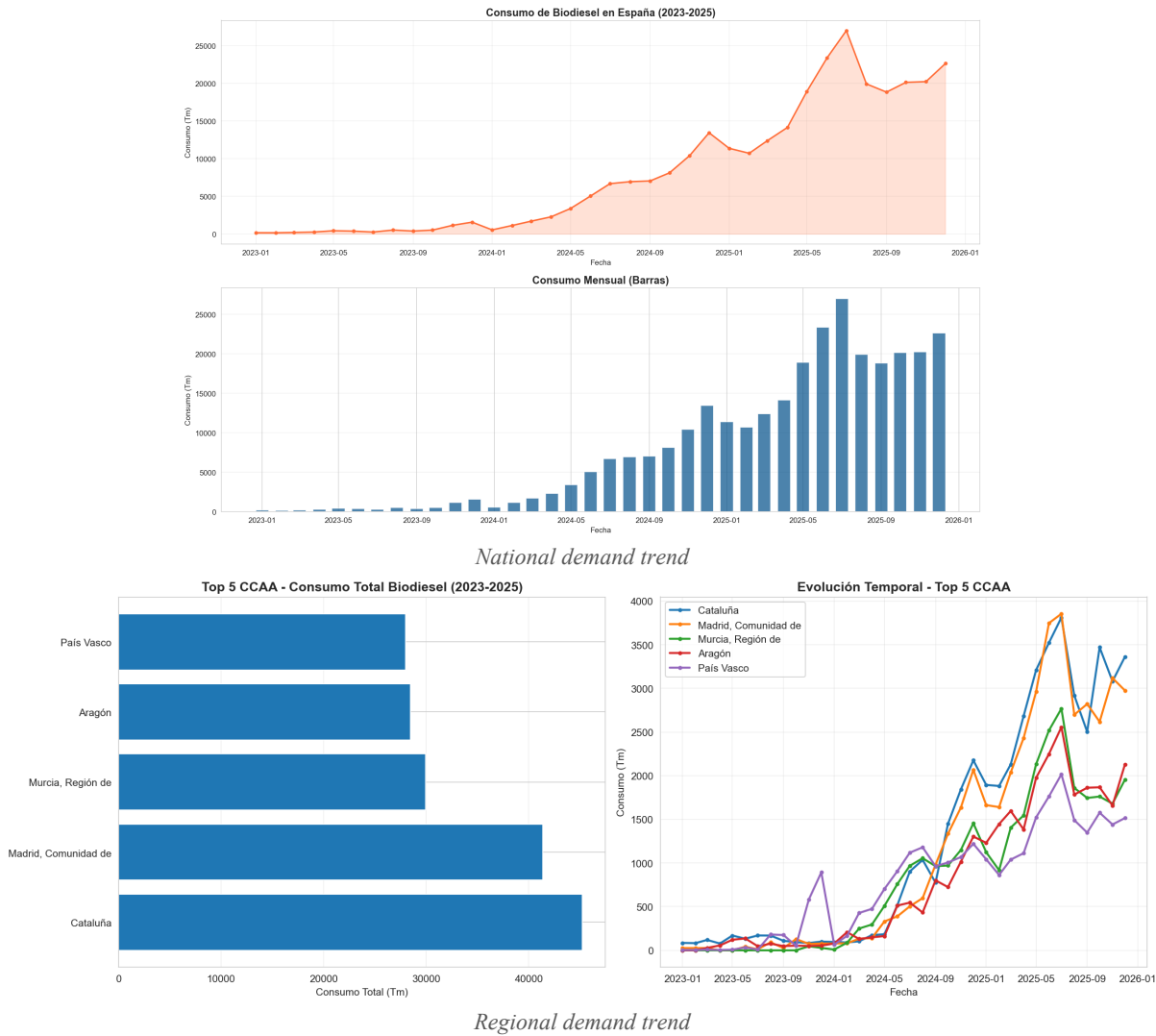
2.8 Modeling Dataset

The feature-engineered modeling dataset is saved as:

`data/features/features_modelo_completo.csv`

Current dimensions:

- 180 rows.
- 35 columns.
- 5 target series.
- Monthly coverage from 2023-01 to 2025-12.



3. Methodology

3.1 Technical Environment

The project uses a Python-based analytical workflow supported by GitHub version control and notebook-based exploration.

Main tools:

- Python for data processing, feature engineering, modeling, and evaluation.
- pandas and NumPy for data manipulation.
- statsmodels for statistical time-series modeling.
- scikit-learn for machine-learning models and evaluation.
- XGBoost for gradient-boosted tree modeling.
- matplotlib / seaborn for visual analysis.
- GitHub for version control.
- Jupyter notebooks for exploratory and explanatory workflow.

3.2 Pipeline Structure

The project pipeline follows a staged structure:

1. Data cleaning.
2. External data acquisition and preparation.
3. Master dataset construction.
4. Feature engineering.
5. Candidate model training.
6. Model selection and validation.
7. Forecast generation.
8. Business interpretation and reporting.

This structure separates data preparation from modeling, reducing the risk of mixing exploratory logic with final outputs.

3.3 Feature Engineering Methodology

Feature engineering converts the master dataset into a target-level modeling table. The main feature groups are:

- Calendar features: year, month, trend, and seasonal encoding.
- Lag features: previous demand values.
- Rolling features: trailing demand averages.
- Macroeconomic features: industrial production, inflation, and unemployment.
- Fuel-price features: gasoline and diesel price signals.
- Energy-context features: Brent and diesel-pool indicators.

A key rule is to prevent look-ahead leakage. Lag and rolling features are constructed so that each forecast only uses information available before the forecasted month.

4. Potential Solutions

4.1 Alternative 1: Single Time-Series Model for Every Region

The first possible solution was to use a single time-series model family, such as SARIMA, for every target series.

Advantages:

- Easy to explain.
- Strong fit for monthly time-series behavior.
- Handles trend and seasonality.
- Requires fewer external assumptions.

Disadvantages:

- May not capture adoption-curve behavior.
- May fail when a region has a structural demand shift.
- Does not use external drivers unless extended to SARIMAX.

This alternative was useful as a baseline but too restrictive for the full project.

4.2 Alternative 2: Machine-Learning Model With External Drivers

The second possible solution was to use machine-learning models such as Random Forest or XGBoost with macroeconomic, fuel-price, and lag features.

Advantages:

- Can capture non-linear relationships.
- Can use many explanatory variables.
- Can identify feature importance.
- Flexible across regions.

Disadvantages:

- Requires more data than available.
- Can struggle to extrapolate outside the training range.
- May overfit short time series.
- Recursive long-horizon forecasting can compound error.

This alternative was technically valuable, but it was not reliable enough to be the only forecasting approach.

4.3 Alternative 3: Growth-Curve Models for Adoption Behavior

The third possible solution was to use growth-curve models such as Logistic and Gompertz models.

Advantages:

- Well-suited for adoption-stage demand.
- Can represent growth that slows over time.
- More realistic than unlimited linear or exponential extrapolation.

Disadvantages:

- Requires assumptions about saturation behavior.
- May underperform if demand has strong seasonality.
- Can be sensitive to the short historical window.

This alternative was important because biodiesel demand behaves like an adoption-stage product in several target markets.

4.4 Chosen Solution

The final solution is a model-comparison framework rather than a single-model approach. The project evaluates multiple candidate models and selects the best usable model separately for each target.

Candidate model families:

- SARIMA
- SARIMAX
- Logistic growth curve
- Gompertz growth curve
- Ridge regression
- Random Forest
- XGBoost

This approach is more appropriate because each region can behave differently. Madrid and the national market may follow adoption-curve dynamics, while Catalonia and Andalusia may be better represented by time-series dynamics.

5. Development

5.1 Repository Workflow

The project is organized into data folders, notebooks, scripts, outputs, and figures. The workflow is designed so that each phase produces an intermediate artifact used by the next phase.

Main folders:

- **data/inputs:** original and consolidated input datasets.
- **data/features:** model-ready feature datasets.
- **data/outputs:** model metrics, forecasts, diagnostics, and scenario outputs.
- **notebooks:** exploratory and explanatory notebooks.
- **scripts:** production-style pipeline scripts.
- **reports/figures:** generated visual outputs.

5.2 Core Development Logic

The pipeline logic can be summarized as:

```
# 1. Load master dataset
df_master = pd.read_csv("data/inputs/master_dataset.csv")

# 2. Filter target rows
df_targets = df_master[df_master["Target"] == 1].copy()

# 3. Build time-series features
df_targets["Lag_1"] = df_targets.groupby("CCAA")["Consumo_Tm"].shift(1)
df_targets["Roll_mean_3"] = (
    df_targets.groupby("CCAA")["Consumo_Tm"]
    .shift(1)
    .rolling(3)
    .mean()
)

# 4. Train candidate models
# 5. Validate with walk-forward logic
# 6. Select the best usable model per target
# 7. Generate 24-month forecasts
```

This code structure emphasizes reproducibility and clarity. Each major output is saved to disk so that results can be audited and reused.

5.3 Model Families Implemented

SARIMA and SARIMAX are used to model temporal structure. Logistic and Gompertz models are used to capture adoption-curve behavior. Ridge, Random Forest, and XGBoost are used as machine-learning baselines with external variables.

The final selected models are:

Target	Final Selected Model	Interpretation
Nacional	Logistic	National demand behaves like an adoption curve.
Madrid	Logistic	Madrid demand is better represented by bounded growth, although error remains high.
Cataluña	SARIMA	Recent time-series dynamics are more useful than growth-curve assumptions.
Andalucía	SARIMA	Recent time-series behavior produces the selected forecast.
Valencia	Gompertz	Valencia appears to retain adoption-stage growth momentum.

5.4 Development Outputs

The main development outputs are:

Output	Path	Description
Master dataset	data/inputs/master_dataset.csv	Unified monthly CCAA dataset.
Feature dataset	data/features/features_modelo_completo.csv	Final modeling feature table.
Selected model metrics	data/outputs/metricas_final_selected.csv	Final holdout metrics by target.
Selected 24-month forecast	data/outputs/forecast_24m_selected.csv	Final selected-model forecasts.
Scenario sensitivity	data/outputs/scenario_sensitivity.csv	Scenario-level forecast sensitivity outputs.
Residual diagnostics	data/outputs/ljung_box_residual_diagnostics.csv	Statistical residual checks.
Safety checks	data/outputs/sarima_safety_check.csv	SARIMA fit diagnostics.

5.5 Forecast Results

The selected 24-month forecast contains 120 rows:

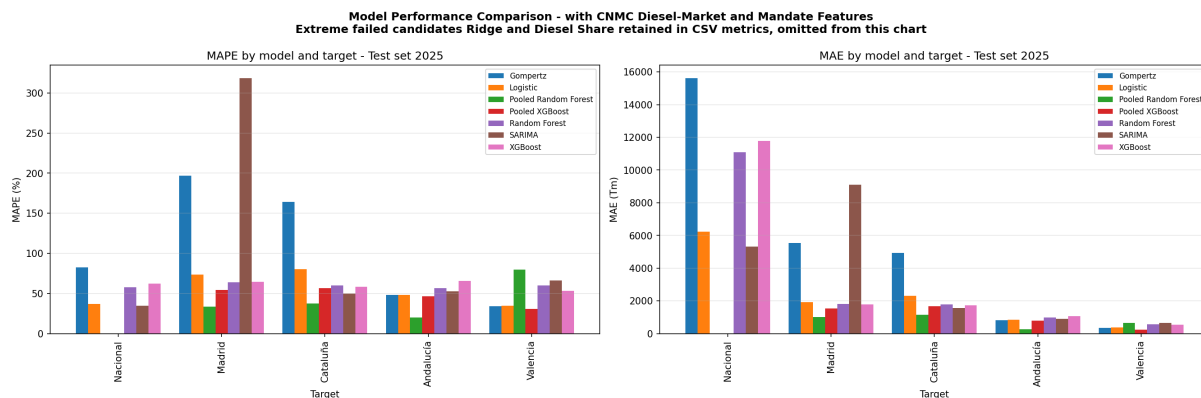
- 5 target series.
- 24 forecast months.
- One selected model per target.

Annual forecast totals:

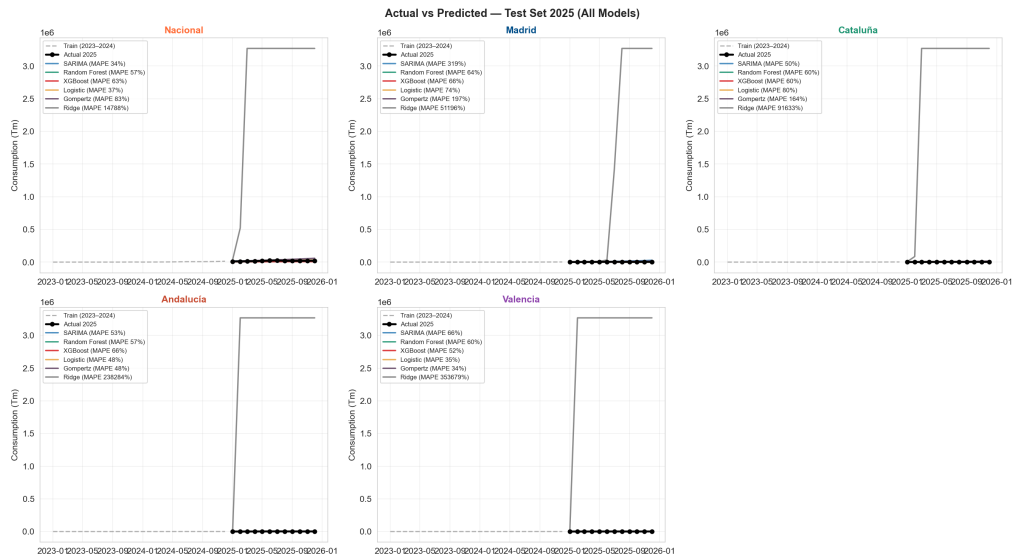
Target	Model	2026 Forecast Tm	2027 Forecast Tm
Nacional	Logistic	310,495.3	323,727.8
Madrid	Logistic	44,848.1	46,007.7
Cataluña	SARIMA	40,654.4	40,778.0
Andalucía	SARIMA	25,434.8	25,046.3
Valencia	Gompertz	17,656.5	20,759.9

Average monthly forecast:

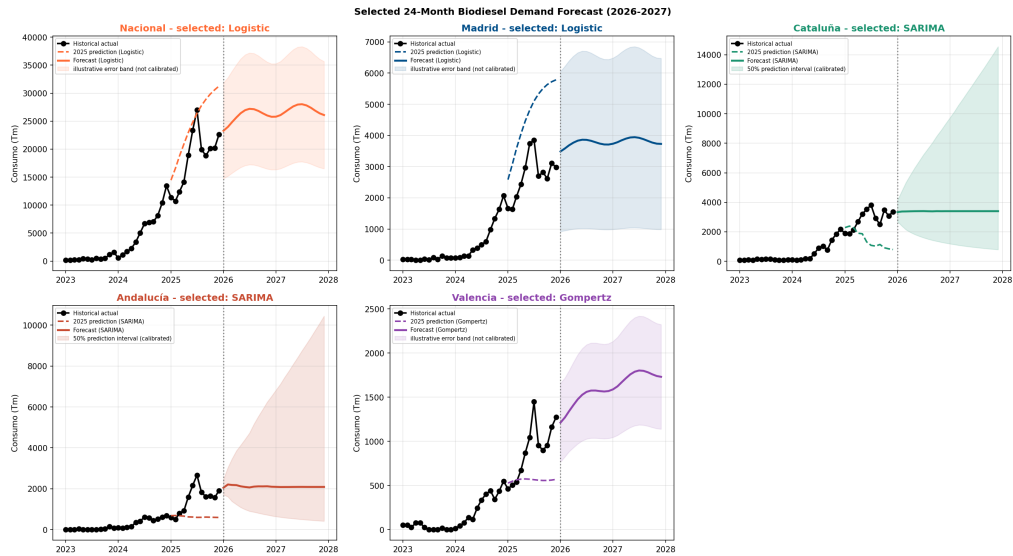
Target	Model	Average Monthly Forecast Tm
Nacional	Logistic	26,426.0
Madrid	Logistic	3,785.7
Cataluña	SARIMA	3,393.0
Andalucía	SARIMA	2,103.4
Valencia	Gompertz	1,600.7



Selected model comparison



Predictions versus actuals



Selected 24-month forecast



Residual diagnostics

5.6 Regional Interpretation

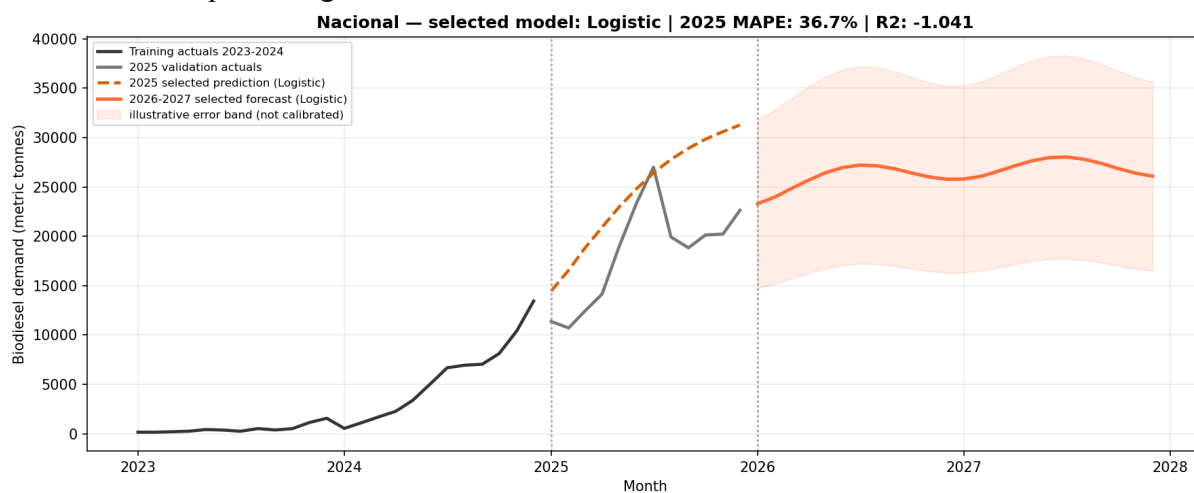
Nacional is selected with a Logistic model. This suggests a national adoption-curve pattern, where demand continues to grow but should not be extrapolated without limits.

Madrid is also selected with a Logistic model, but the holdout error is high. This means the Madrid forecast should be interpreted directionally and validated with additional market information before operational use.

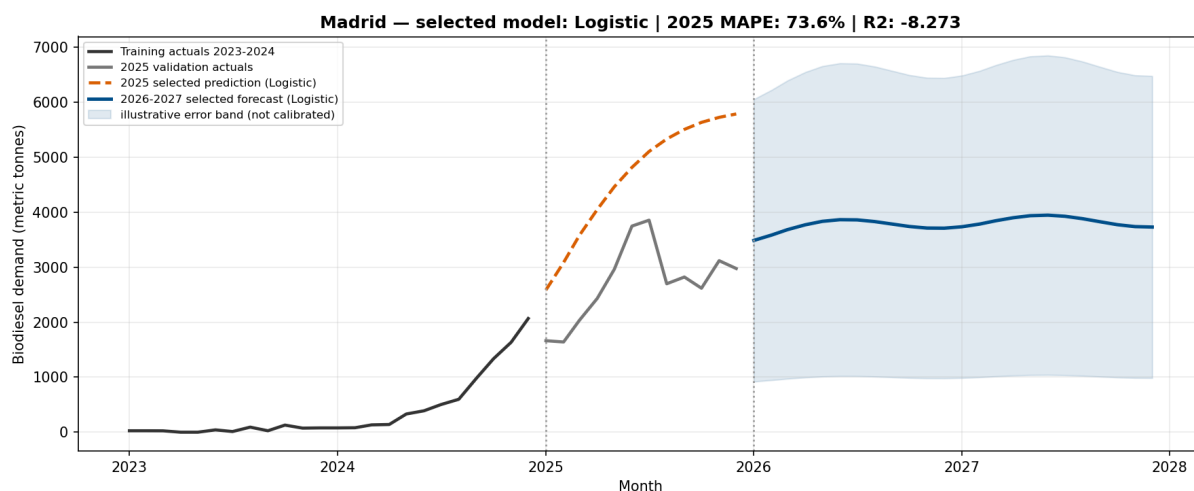
Catalonia is selected with SARIMA. Its forecast is relatively stable, suggesting that recent time-series behavior is more useful than a growth-curve assumption.

Andalusia is selected with SARIMA. The forecast also appears stable, with moderate month-to-month variation.

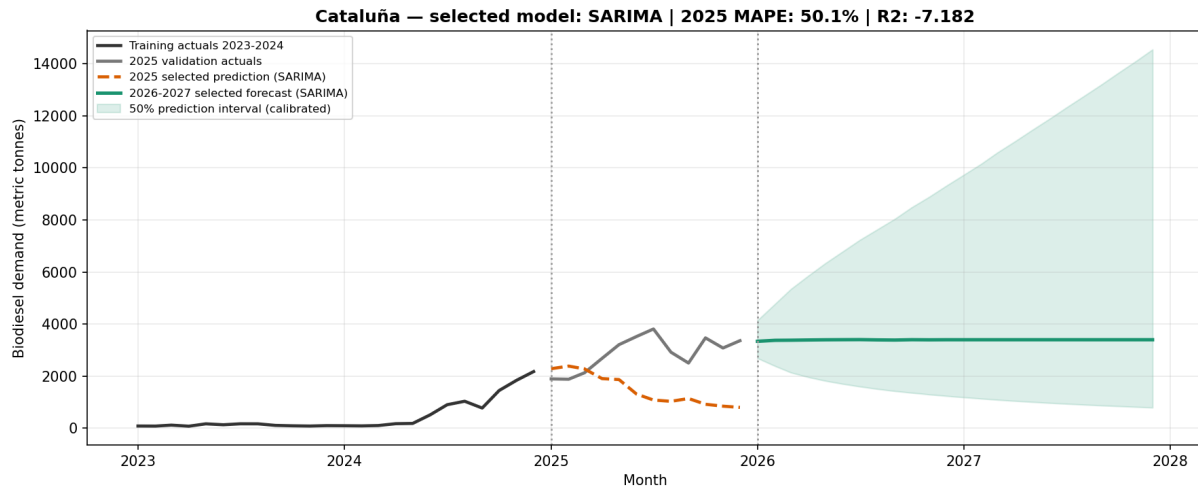
Valencia is selected with Gompertz. This suggests ongoing growth momentum with a shape consistent with adoption-stage demand.



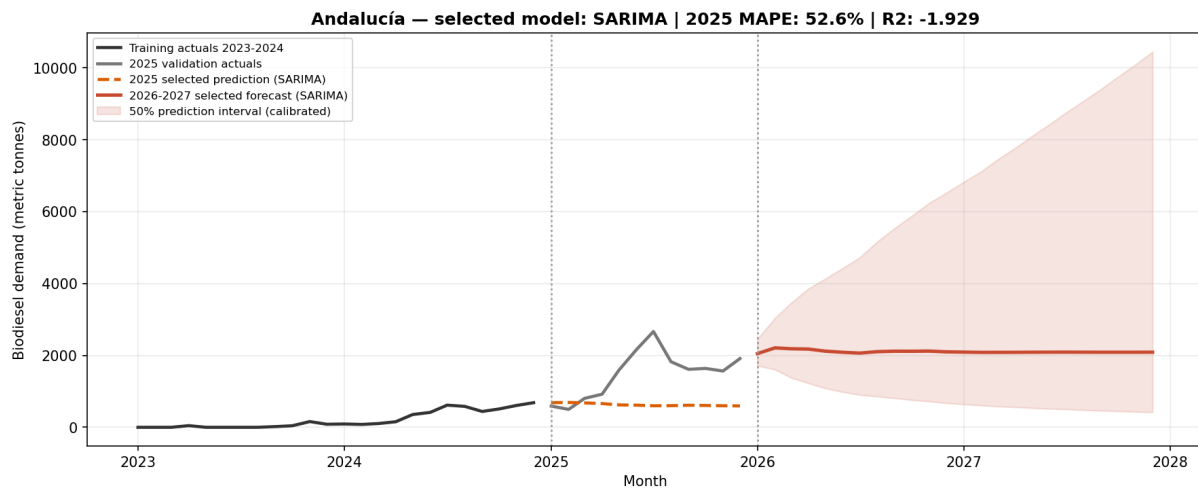
Nacional business forecast



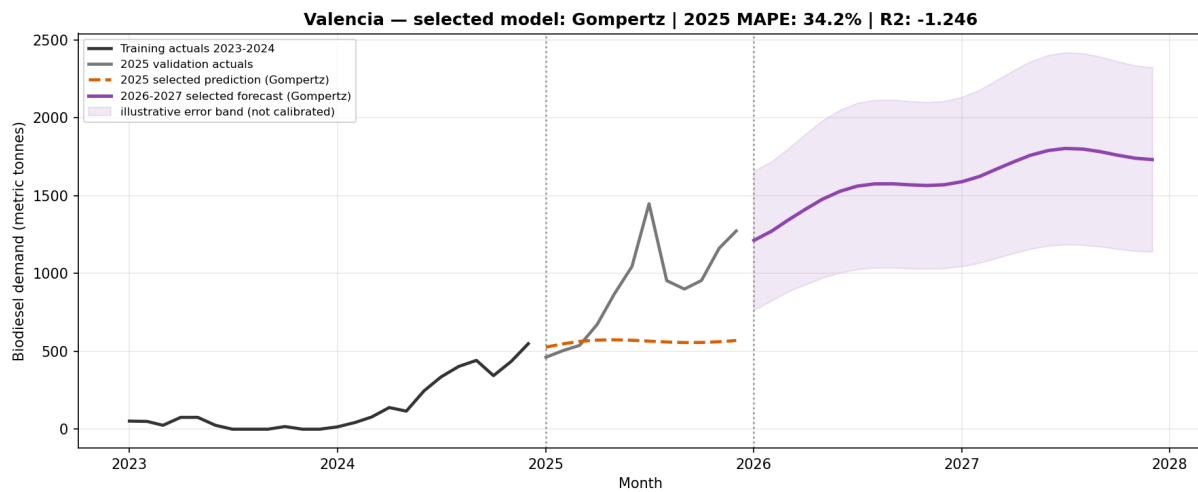
Madrid business forecast



Catalonia business forecast



Andalusia business forecast



Valencia business forecast

6. Conclusions

6.1 Final Technical Result

The project successfully builds a reproducible forecasting pipeline for Diesel Nexa / biodiesel demand. The pipeline consolidates multiple datasets, creates a master dataset, engineers time-series features, evaluates multiple model families, and generates selected 24-month forecasts for national and regional targets.

The final selected models demonstrate that one algorithm is not sufficient for every region. Logistic and Gompertz models are useful where demand behaves like an adoption curve, while SARIMA is useful where recent temporal patterns dominate.

6.2 Technical Performance

The model performance metrics show that forecast uncertainty remains high:

- The best MAPE is Valencia at 34.2%.
- The national model has a MAPE of 36.7%.
- Catalonia and Andalusia have MAPE values above 50%.
- Madrid has the highest MAPE at 73.6%.

These errors reflect the nature of the problem. The target series has only 36 monthly observations and appears to be in a growth/adoption phase. High error is therefore expected and should be communicated transparently.

6.3 Business Impact

The project provides a structured forecasting foundation for Diesel Nexa planning. It allows stakeholders to compare national and regional demand expectations, understand model uncertainty, and identify where further data collection would improve reliability.

The forecast is most useful for:

- Strategic planning.
- Regional prioritization.
- Medium-term supply discussions.
- Scenario thinking.
- Executive communication.

The forecast should not be treated as a final operational planning model without further validation.

6.4 Practicality for Implementation

The current implementation is practical because it is reproducible, transparent, and modular. It can be rerun when new monthly data becomes available. The use of saved intermediate outputs also makes it easier to audit assumptions and update specific steps without rebuilding the entire project manually.

The next stage should focus on improving model reliability by adding more historical observations, regional macroeconomic variables, regulatory event features, and stronger scenario assumptions.

6.5 Final Recommendation

The recommended business use is to treat the current forecasts as a first strategic demand-planning layer. They are valuable for understanding likely direction and relative regional behavior, but they should be combined with expert judgment and additional market information before making binding operational decisions.

Bibliography

Comision Nacional de los Mercados y la Competencia (CNMC). Petroleum product consumption statistics and diesel-market data used to reconcile biodiesel demand and construct diesel-pool context variables.

Corporacion de Reservas Estrategicas de Productos Petroliferos (CORES). Certified biofuel and biodiesel consumption data used as historical and supplementary context for the project demand series.

Instituto Nacional de Estadistica (INE). Macroeconomic indicators used in the project, including industrial production, consumer price variation, and unemployment rate variables.

Fuel-price datasets for Spain, 2023-2025. Regional gasoline and diesel price observations processed into monthly autonomous-community price features.

Brent crude oil monthly price data, 2023 onwards. International crude oil price context used as an energy-market explanatory variable.

Project repository artifacts. `README.md`, `DATA_AUDIT_REPORT.md`, `INDEPENDENT_AUDIT_REPORT.md`, production scripts, notebooks, and generated outputs in the Repsol capstone repository.

Python Software Foundation. Python programming language used for data processing, modeling, and report automation.

pandas, NumPy, statsmodels, scikit-learn, XGBoost, matplotlib, and seaborn documentation. Python libraries used for data manipulation, statistical modeling, machine learning, visualization, and evaluation.

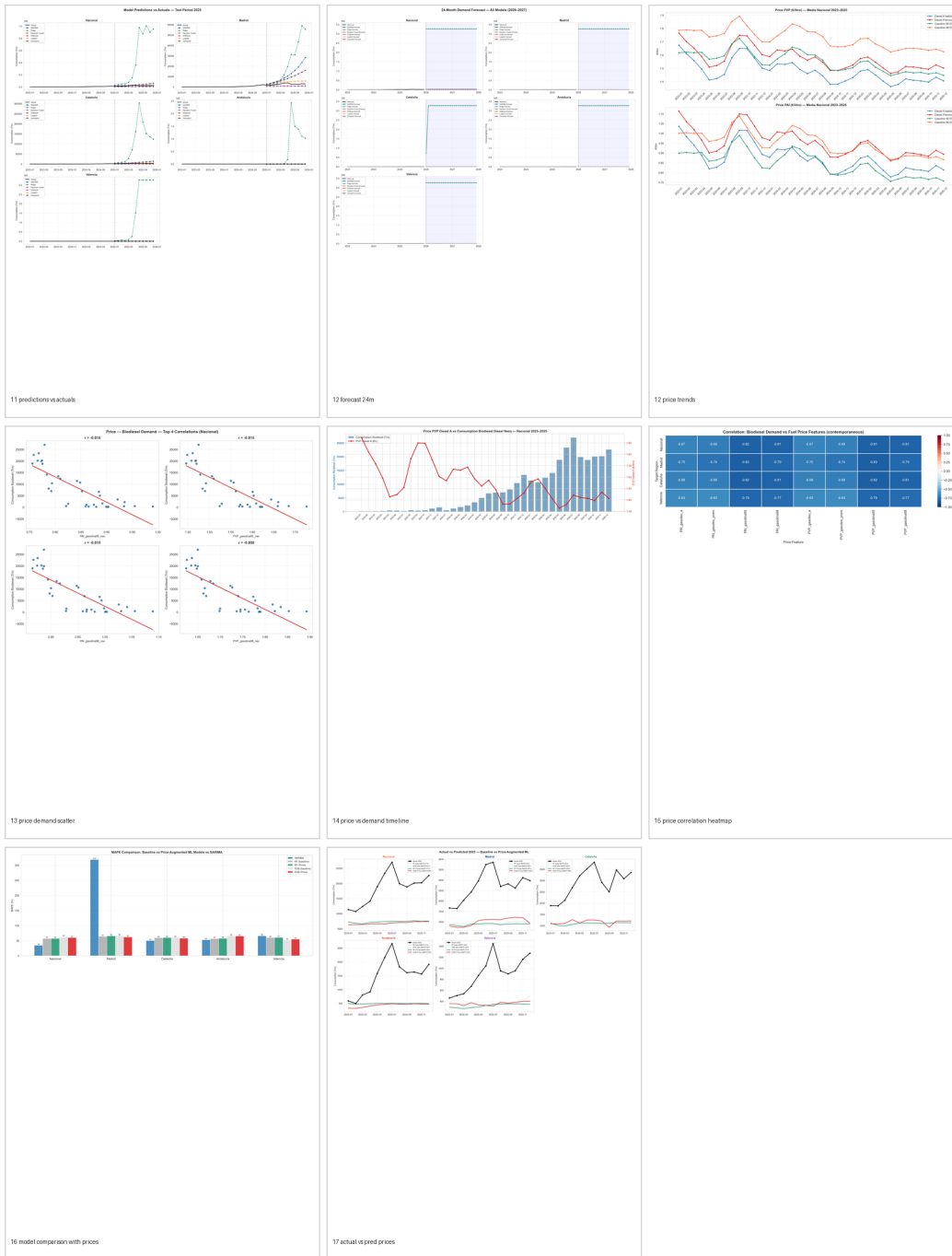
Appendix: Additional EDA and Diagnostic Figures

The following appendix sheets collect additional exploratory and diagnostic charts generated during the project but not included individually in the main body of the report.

Appendix Figures Sheet 1: Additional EDA and Diagnostic Visuals



Appendix figures sheet 1



Appendix figures sheet 2